

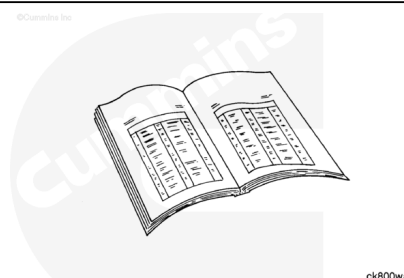
Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

Note : If an air compressor is not mounted on the engine, it will not be necessary to drain the cooling system.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html>)
- Remove the air compressor. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/28/28-012-014-tr.html>)

- Remove the accessory drive pulley. Refer to Procedure 009-004 in Section 9.
(/qs3/pubsys2/xml/en/procedures/28/28-009-004-tr.html)
- Remove the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9.
(/qs3/pubsys2/xml/en/procedures/28/28-009-039-tr.html)
- Remove the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-003-tr.html)

with Electronically Actuated Injector

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ck800wa

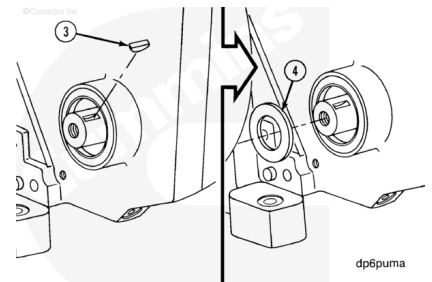
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
- Remove the accessory drive pulley. Refer to Procedure 009-004 in Section 9.
(/qs3/pubsys2/xml/en/procedures/28/28-009-004-tr.html)
- Remove the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9.
(/qs3/pubsys2/xml/en/procedures/28/28-009-039-tr.html)
- Remove the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-003-tr.html)

Remove

with Mechanically Actuated Injector

⚠ CAUTION ⚠

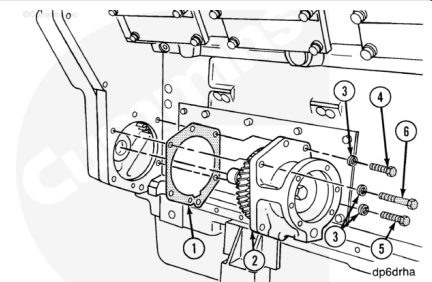
Make sure the woodruff key has been removed before attempting to remove the fuel pump drive assembly. Damage to the bushing in the front gear housing cover can result if the key is not removed.



Remove the oil slinger.

Remove the six capscrews from the drive housing flange.

Remove the drive and discard the gasket.

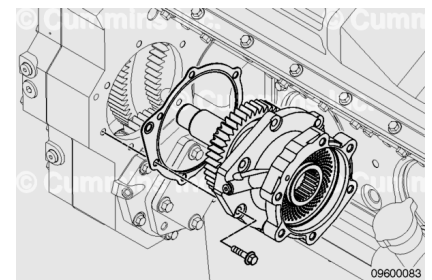


with Electronically Actuated Injector

Remove the six capscrews.

Remove the accessory drive assembly.

Remove and inspect the gasket.

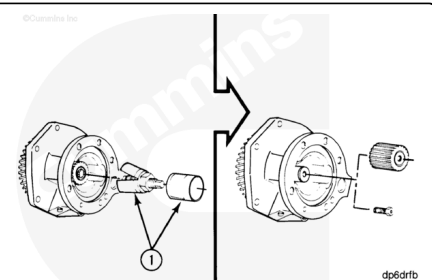


Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

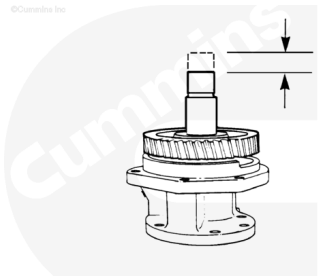


Use solvent to clean the outer housing of the accessory drive.

Use a dial indicator to measure the fuel pump drive shaft end clearance.

Fuel Pump Drive Shaft End Clearance with Mechanically Actuated Injector

mm		in
0.05	MIN	0.002
0.30	MAX	0.012

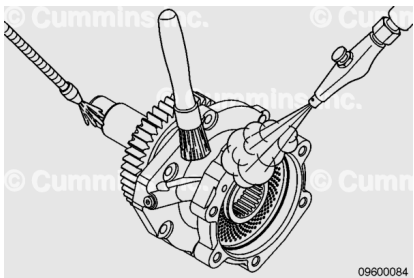


dp6dca

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



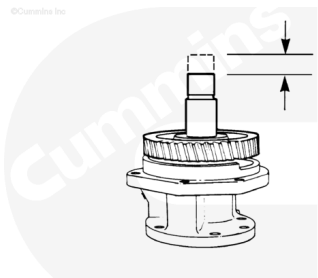
09600084

Use solvent to clean the outer housing of the accessory drive.

Use a dial indicator to measure the fuel pump drive shaft end clearance.

Fuel Pump Drive Shaft End Clearance with Electronically Actuated Injector

mm		in
0.14	MIN	0.006
0.27	MAX	0.011



dp6dca

Disassemble

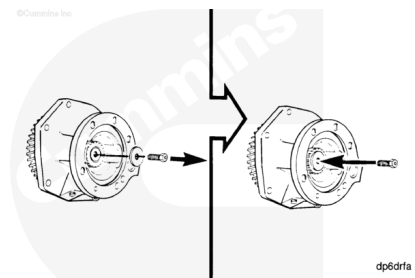
with Mechanically Actuated Injector

⚠ CAUTION ⚠

To prevent damage to the shaft, the capscrew must be installed into the drive shaft without the washer.

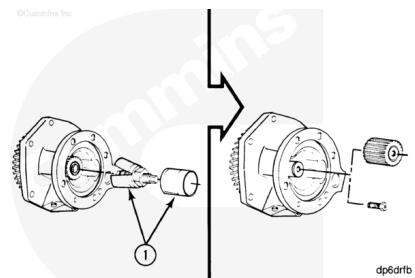
Remove the capscrew and washer.

Install the capscrew into the shaft.



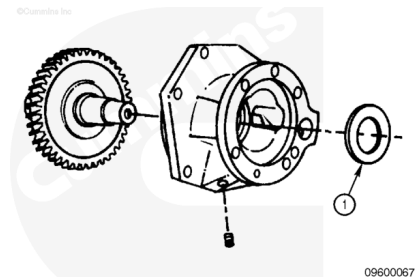
Use a jaw puller to remove the lovejoy or spline type coupling.

Remove the capscrew from the shaft.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Remove the clamping washer (1).

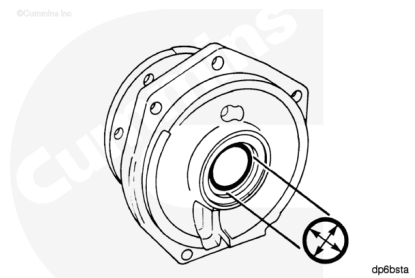
Remove the gear and shaft assembly.

Remove the pipe plug from the housing.

Use solvent to clean the parts.

Measure the inside diameter of the bushings.

Accessory Drive Bushing Inside Diameter		
mm		in
33.566	MIN	1.3215
33.642	MAX	1.3245



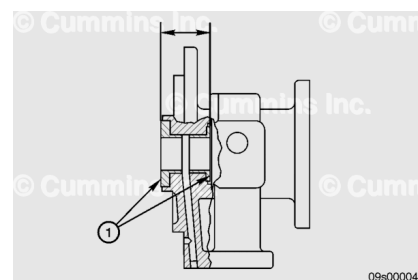
If the inside diameter of either bushing is **not** within specification, both bushings **must** be replaced.

Check the grooved surfaces of the bushings (1) for damage.

Measure the distance from the outside of the thrust faces of the two bushings.

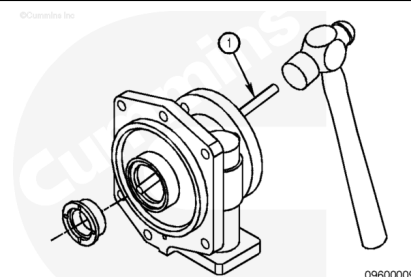
Accessory Drive Bushing Thrust Face Distance

mm		in
45.517	MIN	1.7920
46.644	MAX	1.7970



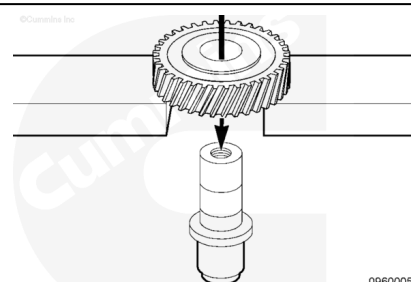
If the distance between the two thrust faces is **not** within specification, both bushings **must** be replaced.

Remove the flanged bushing from the drive housing using a brass drift (1), taking care **not** to damage the bushing bore.



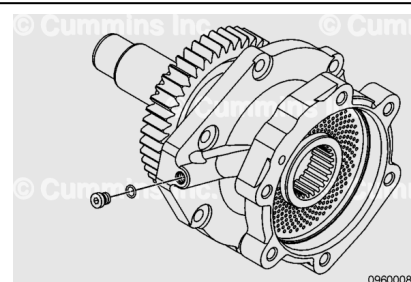
Only remove the gear from the shaft when the gear or the shaft **must** be replaced.

Use an arbor press to remove the gear.

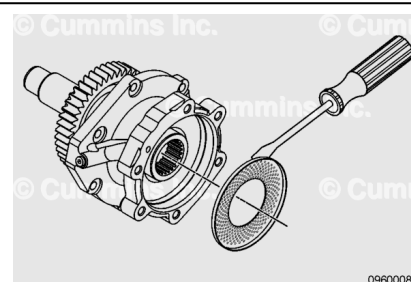


with Electronically Actuated Injector

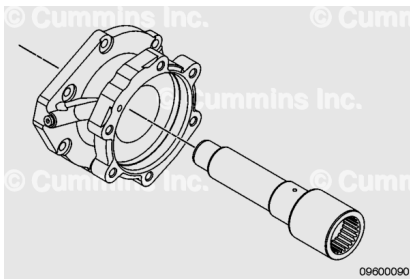
Remove the straight thread o-ring plug from the fuel pump drive.



Remove the mesh screen with a small flat tip screwdriver.

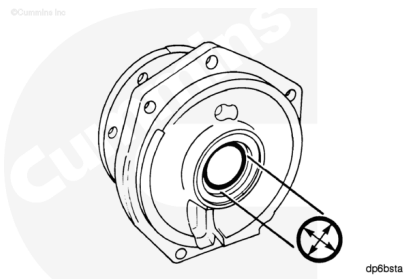


Use a press to remove the shaft from both the gear and the pump drive housing.



Measure the inside diameter of the bushings.

Fuel Pump Drive Bushing Inside Diameter		
mm		in
50.000	MIN	1.9685
50.030	MAX	1.9697

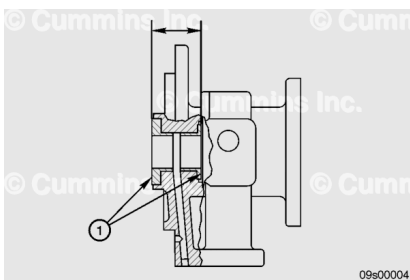


If the inside diameter of either bushing is **not** within specification, both bushings **must** be replaced.

Check the grooved surfaces of the bushings (1) for damage.

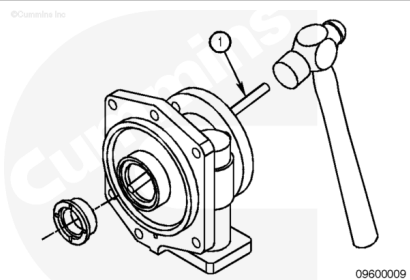
Measure the distance from the outside of the thrust faces of the two bushings.

Fuel Pump Drive Bushing Thrust Face Distance		
mm		in
48.870	MIN	1.9240
48.950	MAX	1.9272



If the distance between the two thrust faces is **not** within specification, both bushings **must** be replaced.

Remove the flanged bushing from the drive housing. Use a brass drift (1), taking care **not** to damage the bushing bore.

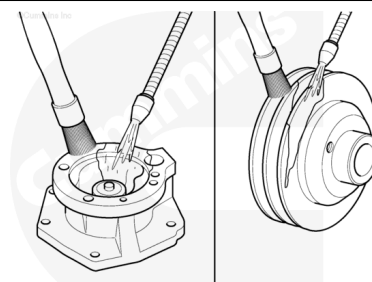


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



08600657

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

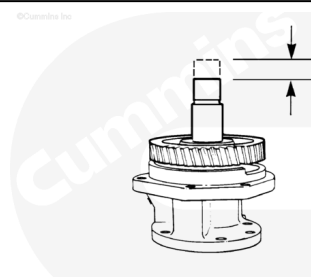
Clean the fuel pump drive assembly and inspect it for reuse.

Clean the pulley and inspect if for reuse, if applicable.

Measure the end clearance.

Fuel Pump Shaft Drive End Clearance with
Mechanically Actuated Injector

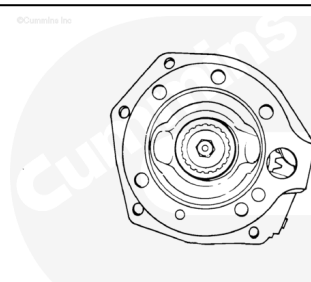
mm		in
0.05	MIN	0.002
0.30	MAX	0.012



dp6drcb

Note : Oversize thrust bearings are available to adjust the end clearance.

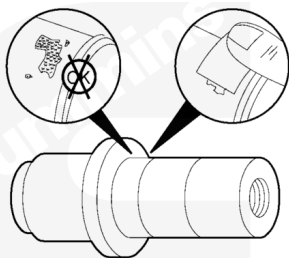
The coupling washer **must** be clamped tightly between the coupling and the shaft. If the washer is **not** tight, rebuild the drive unit.



dp6drcb

Inspect the shaft in the gear fit area for fretting or burrs.

Clean burrs with emery cloth.



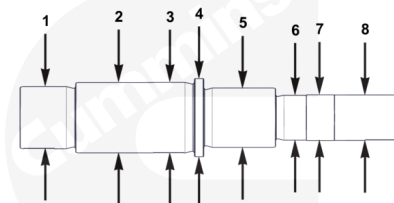
09600059

Measure the shaft outside diameter.

Accessory Drive Shaft Outside Diameter

	mm		in
(1)	34.9631	MIN	1.3765
	34.9758	MAX	1.3770
(2)	39.6621	MIN	1.5615
	39.6748	MAX	1.5620
(3)	39.7891	MIN	1.5665
	39.8018	MAX	1.5670
(4)	43.942	MIN	1.730
	44.196	MAX	1.740
(5)	33.3121	MIN	1.3115
	33.3248	MAX	1.3120
(6)	25.019	MIN	0.985
	25.273	MAX	0.995
(7)	25.5270	MIN	1.0050
	25.5422	MAX	1.0056
(8)	25.4254	MIN	1.0010
	25.4406	MAX	1.0016

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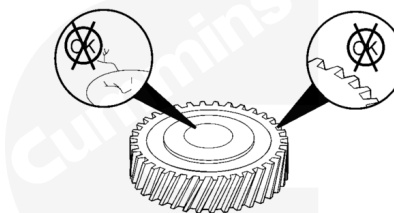
09s00018

If the outside diameter is **not** within specification, the shaft **must** be replaced.

Check the gear teeth and gear bore for damage.

Replace the gear, if damaged.

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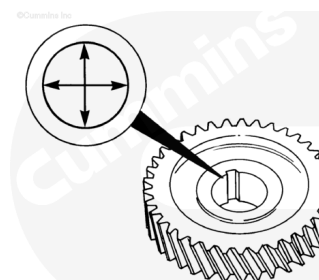


09600057

Measure the gear inside diameter.

Accessory Drive Gear Inside Diameter

mm		in
44.98	MIN	1.771
45.02	MAX	1.772



dp1gela

If the gear is **not** within specification, the gear **must** be replaced.

Check the inner diameter of the coupling for damage.

If damage is found the coupling **must** be replaced.

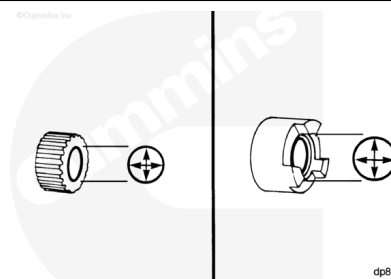
Measure the inside diameter of the coupling.

Accessory Drive Lovejoy Coupling Inside Diameter

mm		in
25.377	MIN	0.999
25.407	MAX	1.000

Accessory Drive Spline Coupling Inside Diameter

mm		in
25.255	MIN	0.994
25.278	MAX	0.996

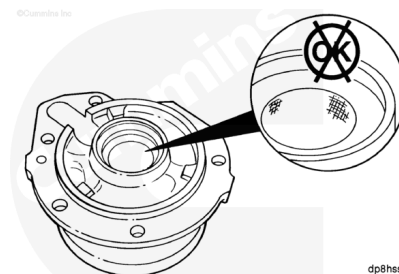


dp8cpta

If the inner diameter of the coupling is **not** within specification the coupling **must** be replaced.

If the bushing was removed, inspect the housing for scoring and other damage.

Replace the housing if damaged.

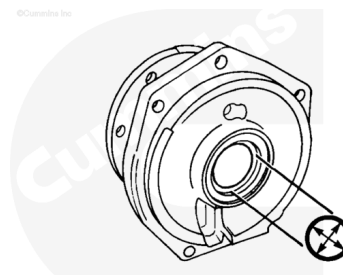


dp8hssa

If the bushing was removed, measure the inside diameter of the bushing bore.

Accessory Drive Bushing Bore Inside Diameter

mm		in
39.611	MIN	1.5595
39.637	MAX	1.5605



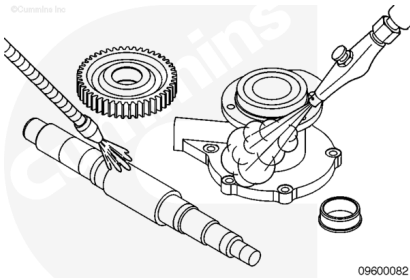
09600017

Replace the housing if the bushing bore is **not** within specifications.

with Electronically Actuated Injector

⚠ WARNING ⚠

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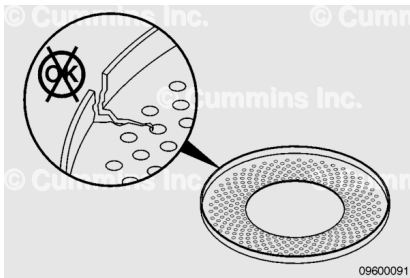


⚠ WARNING ⚠

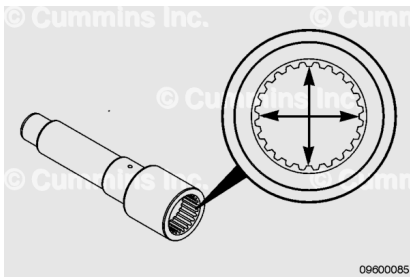
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the fuel pump drive components.
Dry with compressed air.

Inspect the screen for cracks or other damage.
If any damage is found, the screen **must** be replaced.



Check the inner diameter of the shaft spline for damage.
If damage is found, the shaft **must** be replaced.
Measure the inside diameter of the shaft.

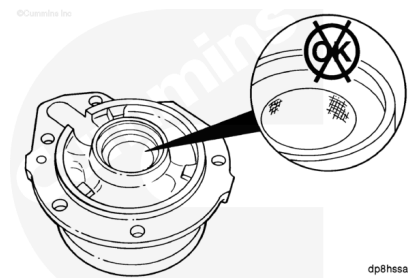


Fuel Pump Drive Spline Inside Diameter		
mm		in
46.000	MIN	1.811
46.190	MAX	1.819

If the inner diameter of the shaft is **not** within specification, the shaft **must** be replaced.

If the bushing was removed, inspect the housing for scoring and other damage.

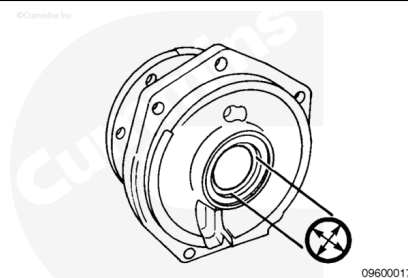
Replace the housing, if damaged.



If the bushing was removed, measure the inside diameter of the bushing bore.

Fuel Pump Drive Bushing Bore Inside Diameter

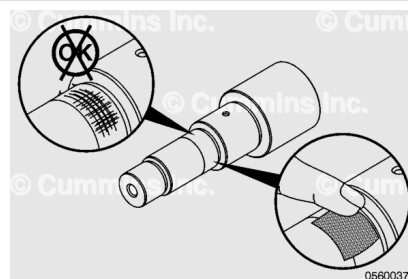
mm		in
60.000	MIN	2.3622
60.030	MAX	2.3634



Replace the housing if the bushing bore is **not** within specifications.

Inspect the shaft in the gear fit area for fretting or burrs.

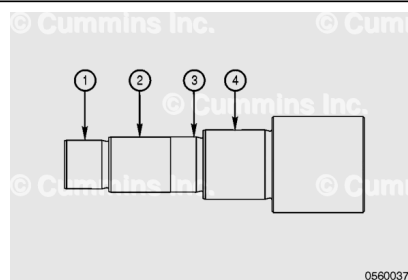
Clean burrs with emery cloth.



Measure the shaft outside diameter.

Fuel Pump Drive Shaft Outside Diameter

	mm		in
(1)	34.963	MIN	1.376
	34.978	MAX	1.377
(2)	39.662	MIN	1.561
	39.674	MAX	1.562
(3)	39.789	MIN	1.566
	39.801	MAX	1.567
(4)	49.820	MIN	1.961
	49.832	MAX	1.962

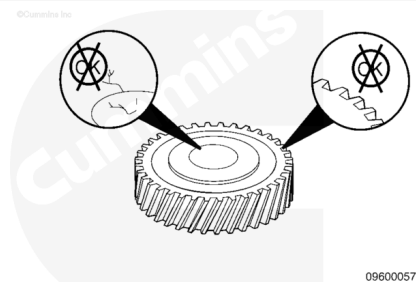


If the outside diameter is **not** within specification, the shaft **must** be replaced.

Note : Dimensions 2 and 3 are on the same section of the shaft. Dimension 3 **only** applies in the area adjacent to the shoulder, where the keyway is located.

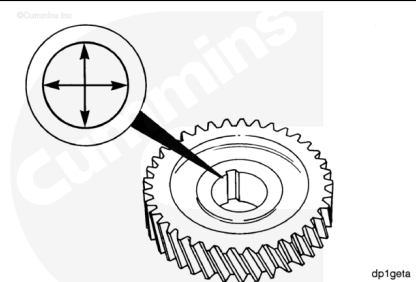
Check the gear teeth and gear bore for damage.

Replace the gear, if damaged.



Measure the gear inside diameter.

Fuel Pump Drive Gear Inside Diameter		
mm		in
39.726	MIN	1.564
39.751	MAX	1.565



If the gear is **not** within specification, the gear **must** be replaced.

Assemble

with Mechanically Actuated Injector

⚠ WARNING ⚠

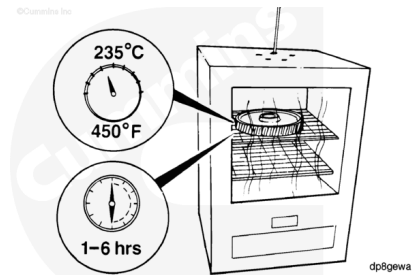
Wear protective clothing to reduce the possibility of personal injury from burns.

⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the gear teeth will result.

If an adequate press is **not** available, an oven can be used.

Heat the gear at 235°C [450°F] for **not** less than one hour, and **not** more than six hours.

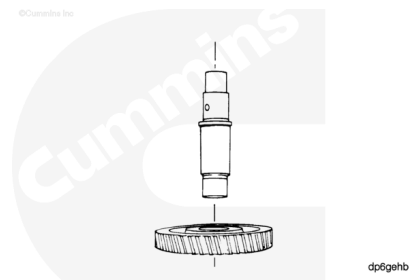


⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

Support the gear.

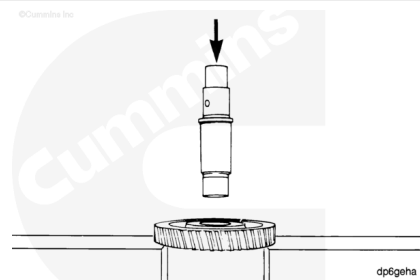
Slide the shaft into the gear until the shoulder of the shaft touches the gear.



If an arbor press is available, support the gear.

Lubricate the shaft with clean engine oil.

Press the shaft through the gear until the shoulder of the shaft touches the gear.



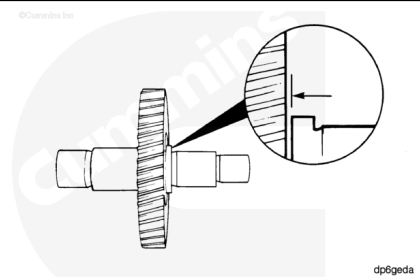
⚠ CAUTION ⚠

If the heat method was used to install the gear, allow the gear to air cool. Do not use water or oil to reduce cooling time. Damage to the gear can result.

Use a feeler gauge to measure the distance between the shoulder of the shaft and the gear.

Accessory Drive Gear to Shaft Distance		
mm		in
0.05	MAX	0.002

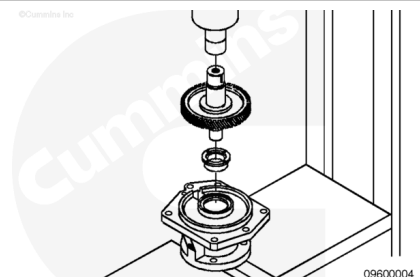
If the distance between the gear and the shaft is **not** within specification, press the gear on until the specification is met.



⚠ CAUTION ⚠

The bushings must be pressed in square to the bores of the housing. The thrust faces of the two bushings must be parallel and concentric. The bushing with the center counterbored must have the pressure applied to the counterbore face and not on the thrust bearing face. Failure to comply can result in distortion of the thrust face.

Support the housing.



The bushing with the smaller outside diameter is pressed into the output side.

Using an arbor press and the shaft as a mandrel, press the bushing into the housing.

Turn the housing over and press the remaining bushing into place.

Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105, Part Number 3163086, 3163087 or equivalent.

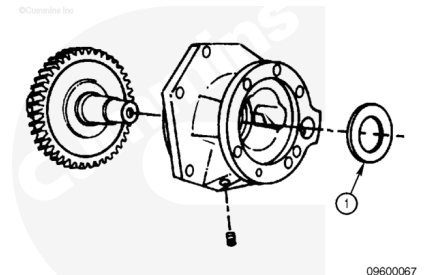
Install the gear and shaft assembly into the housing.

Install the clamping washer (1) with the beveled edge up.

Install the pipe plug into the housing.

Tighten the pipe plug.

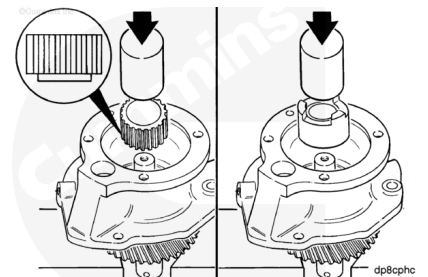
Torque Value: 8 n•m [75 in-lb]



Support the assembly on the gear or the shaft.

Use an arbor press and a mandrel to press the coupling onto the shaft until it touches the clamping washer.

The clamping washer **must** be positioned tightly between the coupling and the shoulder of the shaft.



The capscrew **must** contain an oil drilling if an air compressor is to be mounted on the engine.

Install the washer and capscrew.

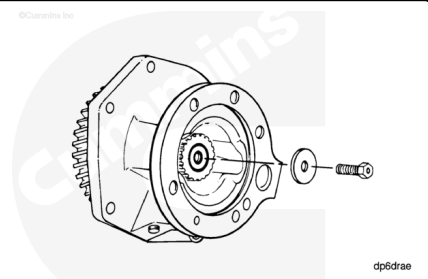
Tighten the capscrew.

Torque Value:

9.525 mm [0.375 in] Capscrew Length 45 n•m [35 ft-lb]

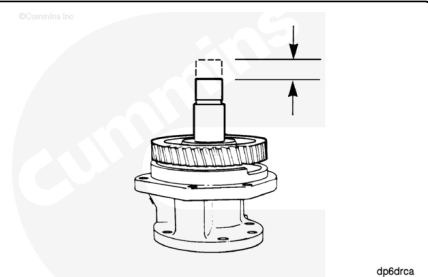
Torque Value:

12.7 mm [0.50 in] Capscrew Length 100 n•m [75 ft-lb]



Rotate the shaft, checking for correct assembly.

Use a dial indicator to measure the fuel pump drive shaft end clearance.



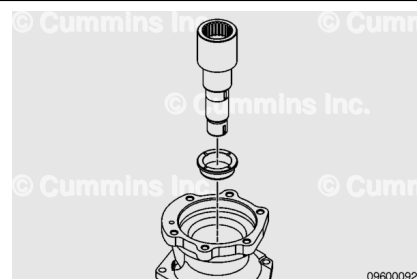
Fuel Pump Drive Shaft End Clearance with Mechanically Actuated Injector

mm		in
0.05	MIN	0.002
0.30	MAX	0.012

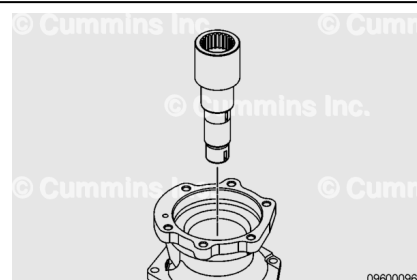
If the end clearance is **not** within specifications, make sure the coupling is positioned tightly against the clamping washer.

with Electronically Actuated Injector

If the bushings were removed, use the shaft as a mandrel and press one bushing into the housing until the underside of the bushing head is flush with the counterbore in the housing. Turn the housing over and press the remaining bushing into place.



Install the shaft into the housing.

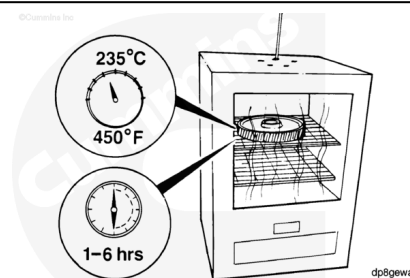


⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the gear teeth will result.

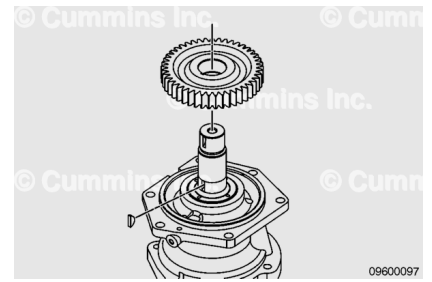


If an adequate press is **not** available, an oven can be used.

Heat the gear to 204°C [400°F] for **not** less than 1 hour, and **not** more than 6 hours.

⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.



Install the key onto the shaft.

Support the shaft and housing.

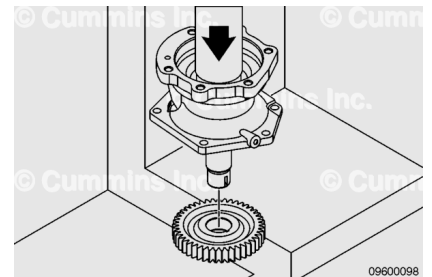
Slide the gear onto the shaft until the shoulder of the shaft touches the gear.

If an arbor press is available, support the gear.

Install the key onto the shaft.

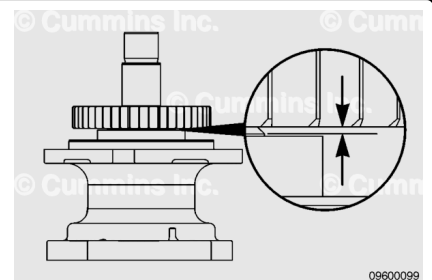
Lubricate the shaft with clean engine oil.

Press the shaft through the gear until the shoulder of the shaft touches the gear.



⚠ CAUTION ⚠

If the heat method was used to install the gear, allow the gear to air cool. Do not use water or oil to reduce cooling time. Damage to the gear can result.

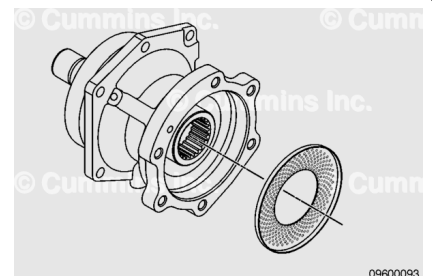


Use a feeler gauge to measure the distance between the shoulder of the shaft and the gear.

Accessory Drive Gear to Shaft Distance		
mm		in
0.05	MAX	0.002

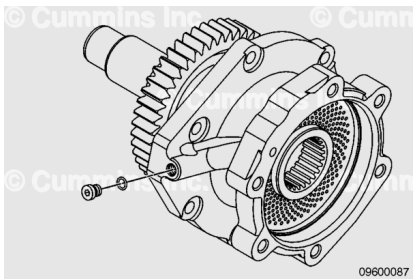
If the distance between the gear and the shaft is **not** within specification, press the gear on until the specification is met.

Install the screen into the housing by pushing the screen in by hand.



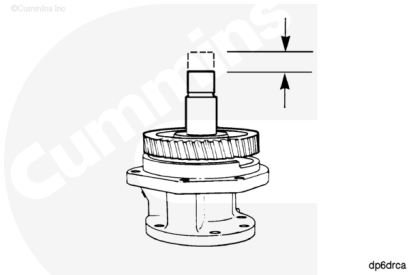
Install the straight thread o-ring plug.

Torque Value: 6 n•m [55 in-lb]



Rotate the shaft, checking for correct assembly.

Use a dial indicator to measure the fuel pump drive shaft end clearance.



Fuel Pump Drive Shaft End Clearance with
Electronically Actuated Injector

mm		in
0.14	MIN	0.006
0.27	MAX	0.011

If the end clearance is **not** within specifications, make sure the coupling is positioned tightly against the clamping washer.

Install

with Mechanically Actuated Injector

Note : This procedure also applies to the drive installed on engines that have a governor drive.

Note : There is no requirement to align the index on the drive gear with the marks on the camshaft gear on the K38 and K50 engines.

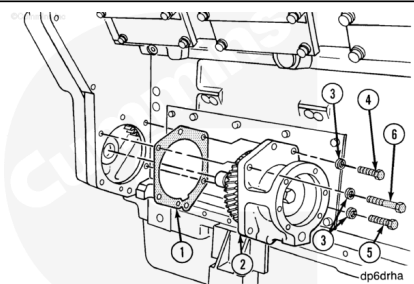
Use Lubriplate™ to lubricate the bushing in the front gear housing cover.

Note : Do not use sealants on the gasket. The gasket is manufactured from the material that sealants will damage.

Install the gasket (1) onto the pilot of the housing cover.

Install the drive assembly (2), turning the shaft as necessary to engage the gears.

Capscrew Specifications:



1. 7/16-14 x 1 inch (one each)
2. 7/16-14 x 1 1/4 inch (one each)
3. 7/16-14 x 3 3/4 inch (one each)

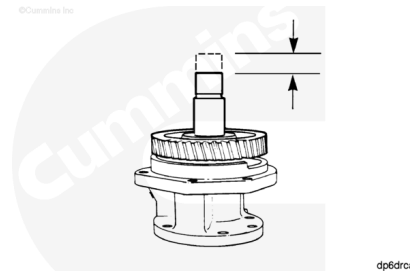
Tighten the capscrews alternately and evenly.

Torque Value: 60 n•m [40 ft-lb]

With the fuel pump drive installed, measure the end clearance.

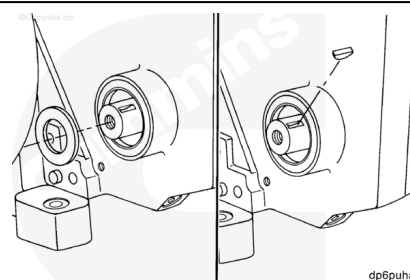
Fuel Pump Drive Shaft End Clearance with
Mechanically Actuated Injector

mm		in
0.05	MIN	0.002
0.30	MAX	0.012



Install the oil slinger.

Install the Woodruff key.



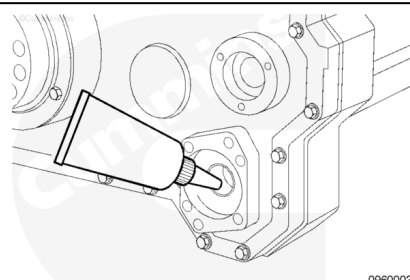
with Electronically Actuated Injector

⚠ CAUTION ⚠

Do not use sealants on the gasket. The gasket is manufactured from material that sealants will damage.

Lubricate the bushing in the front gear housing cover with Lubriplate™ 105.

Install the gasket onto the pilot of the drive housing cover.

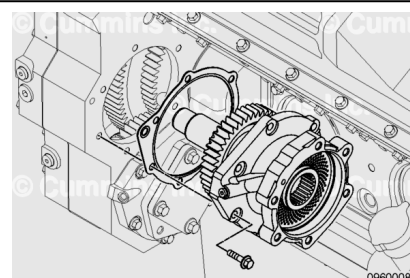


There is **not** a requirement to align the index on the drive gear with the marks on the camshaft gear on QSK38 and QSK50 engines.

Install the fuel pump drive assembly, turning the shaft as necessary to engage the gears.

Install and tighten the six capscrews.

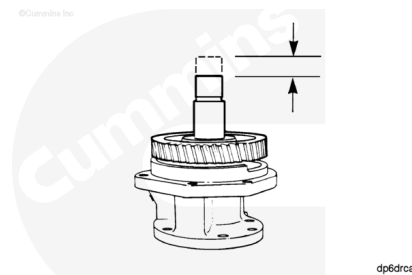
Torque Value: 45 n•m [33 ft-lb]



With the fuel pump drive installed, measure the end clearance.

Fuel Pump Drive Shaft End Clearance with
Electronically Actuated Injector

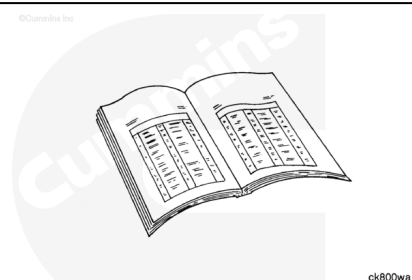
mm		in
0.14	MIN	0.006
0.27	MAX	0.011



Finishing Steps

with Mechanically Actuated Injector

- Install the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-003-tr.html>)
- Install the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9. (</qs3/pubsys2/xml/en/procedures/28/28-009-004-tr.html>)
- Install the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9. (</qs3/pubsys2/xml/en/procedures/28/28-009-039-tr.html>)
- Install the air compressor. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/28/28-012-014-tr.html>)
- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Operate the engine until the coolant temperature reaches 71°C [160°F]. Check for coolant leaks.



with Electronically Actuated Injector

- Install the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-003-tr.html)

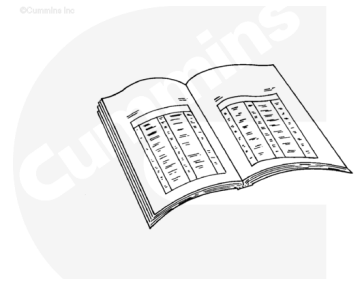
- Install the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9.

(/qs3/pubsys2/xml/en/procedures/28/28-009-004-tr.html)

- Install the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9.

(/qs3/pubsys2/xml/en/procedures/28/28-009-039-tr.html)

- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/28/28-005-016-tr.html)
- Operate the engine until the coolant temperature reaches 71°C [160°F] and check for leaks.



ck800wa